1. Please translate the following C function to an assembly subroutine.

```
int32_t find_median(int32_t *pArray, int32_t size){
    if(size % 2 == 0){
        return (pArray[size/2] + pArray[size/2 + 1])/2;
    }else{
        return pArray[size/2 + 1];
    }
}
```

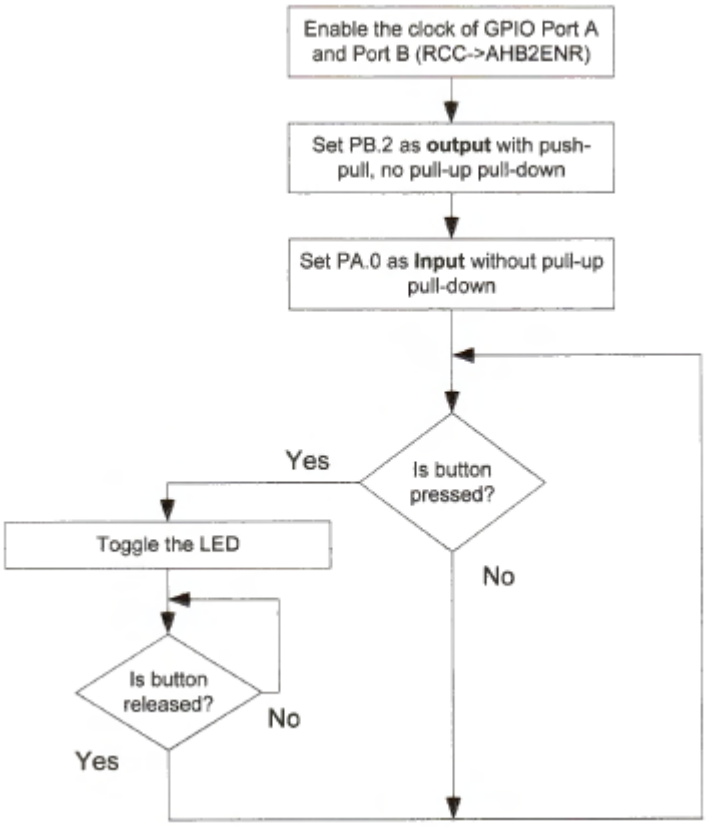
2. Please translate the following assembly subroutine to a C function.

```
myfun PROC
EXPORT myfun
VMUL.F32 s0, s0, s0
VLMA.F32 s0, s1, s1
VLMA.F32 s0, s2, s2
VSQRT.F32 s0, s0
BX lr
ENDP
```

3. Suppose a timer has the following setting
   Input clock=16Mhz
   PSC=159
   ARR=1999
   CCR=499
   Direction=Up-counting

   a. Calculate the PWM frequency.
   b. Calculate the duty cycle in low-true mode.
   c. Calculate the duty cycle in high-true mode.

4. Suppose a push btn is connected to PA.0 and an LED is attached to PB.2. Please implement the C code for the program shown in the flowchart.



5. In the following timer interrupt handler r3 is used as the overflow counter. However, the main function can't observe any changes in r3 even though overflow events have occurred. Explain why and show your solution to fix this problem.

```
; r3=overflow counter
TIM4_IRQHandlerPROC
EXPORT TIM4_IRQHandler
LDR r0, =TIM4_BASE
LDR r1, [r0, #TIM_SR]
AND r2, r1
CBZ r2, check_CCIF
ADDS r3,r3,#1
check_CCIF:
...
ENDP
```